

FTM 2 MODULE OBJECTIVES				
Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 27	GNET	Population Genetics	SOM.MKIII.BPM1.1.FTM.3.GNET.0086	Analyze how alleles segregate in populations and how to relate allele frequency in populations to genotype frequency of the people in the population
			SOM.MKIII.BPM1.1.FTM.3.GNET.0087	Apply the Hardy Weinberg equation to clinical scenarios to allow estimation of carrier risk, recurrence risk, and frequency of disease alleles in a population for AR, AD, and X-linked genes.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0088	Estimate a carrier frequency from the Hardy Weinberg equation and population incidence for one individual to estimate recurrence risk when that person mates with an individual who has familial carrier risk.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0089	Use Hardy-Weinberg equation to demonstrate why the expected frequency of rare autosomal dominant alleles are expected to be extremely infrequent.
DLA 16	GNET	Population Genetics, Genotype & Allele Frequency, Drift, Founder Effect	SOM.MKIII.BPM1.1.FTM.3.GNET.0082	Describe genetic drift
			SOM.MKIII.BPM1.1.FTM.3.GNET.0083	Discuss founder effect
			SOM.MKIII.BPM1.1.FTM.3.GNET.0084	Explain the meaning of heterozygote advantage
			SOM.MKIII.BPM1.1.FTM.3.GNET.0085	Discuss how new mutation keeps incident rates stable if affected individuals have a lower genetic fitness
DLA 17	BCHM	Proteoglycans and Glycoproteins	SOM.MK.I.BPM1.1.FTM.3.BCHM.0143	Describe the general structure of proteoglycans. Indicate sugar derivatives used for GAG synthesis.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0144	Discuss the general composition and function of GAGs. Describe the bottle-brush structure of proteoglycans.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0145	Describe the synthesis of proteoglycans, hyaluronic acid and the extracellular assembly of proteoglycan aggregates using link proteins leading to shock absorbing aggregates.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0146	Discuss the functions of hyaluronic acid and heparin.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0147	Differentiate O-linked and N-linked glycoproteins and indicate the amino acids involved.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0148	Discuss blood group antigens as an example for O-linked glycoproteins.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0149	Outline synthesis of N-linked glycoproteins and indicate significance of dolichol-PP and mannose.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0150	Highlight the importance of phosphorylation of mannose (Mannose 6-P marker) in sorting of lysosomal enzymes.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0151	Predict the molecular and clinical effects of I-cell disease
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0152	Compare concepts of synthesis of O-linked and N-linked glycoproteins.
Lecture 28	BCHM	Fibrous Proteins	SOM.MK.I.BPM1.1.FTM.3.BCHM.0153	Describe the amino acid composition, structure and function of collagen. Discuss changes in collagen structure in vitamin C deficiency (scurvy).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0154	Discuss the synthesis of procollagen in fibroblasts and its release into the ECM.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0155	Describe the role of lysyl oxidase (copper containing) in covalent cross-linking between tropocollagen molecules in maintenance of collagen stability.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0156	Describe the molecular and genetic basis of Marfan syndrome.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0157	Discuss the molecular basis for occurrence of fractures and presence of blue sclerae in osteogenesis imperfecta (OI).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0158	Describe the amino acid composition, structure and function of elastin and desmosine and highlight significance of lysyl oxidase.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0159	Describe the molecular basis of Ehlers-Danlos Syndrome (EDS).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0160	Compare and contrast the effect of deficiency of prolyl hydroxylase and lysyl oxidase. Compare and contrast the effects of vitamin C deficiency and copper deficiency.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0161	Differentiate Ehlers-Danlos Syndrome (EDS), osteogenesis imperfecta (OI) and Marfan syndrome based on molecular defect and clinical features
Reading Assignment	HCB	Connective Tissue	SOM.MK.I.BPM1.1.FTM.3.HCB.0856	Distinguish between the resident cell types found in connective tissue (fibroblasts, myofibroblasts, macrophages, adipocytes, stem cells and mast cells) based on their microanatomical features and functions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0853	List the function(s) of the different types of connective tissue
			SOM.MK.I.BPM1.1.FTM.3.HCB.0854	List the major components of connective tissue
			SOM.MK.I.BPM1.1.FTM.3.HCB.0855	Distinguish between the wandering cell types found in connective tissue (lymphocyte, plasma cell, neutrophil, monocyte, basophil and eosinophil) based on their location and functions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0857	Describe the role of connective tissue cells in anaphylactic shock.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0858	Microscopically describe the two components that make up the extracellular matrix of connective tissue.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0859	Define ground substance.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0860	Describe the structures of the three main components of the ground substance (proteoglycans, glycosaminoglycans, multiadhesive glycoproteins).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0861	Give at least 2 examples of each main component of the ground substance.

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Lecture 29	HCB	Connective Tissue	SOM.MK.1.BPM1.1.FTM.3.HCB.o862	Describe the cell type responsible for the production of the ground substance.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o863	List the cell types responsible for the production of collagen, elastic and reticular fibers.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o864	Distinguish between collagen, elastic and reticular fibers based on their basic microscopic structures and their function(s).
			SOM.MK.1.BPM1.1.FTM.3.HCB.o865	Identify collagen, elastic and reticular fibers of connective tissue based on their histological staining properties.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o866	List common locations of types I, II and III collagen and elastic fibers.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o867	Define “argyrophillic” and state the fiber of connective tissue that exhibit this property.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o868	List the common types of fibrillar collagens.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o869	List the common types of basement membrane collagens.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o870	Describe the alterations in collagen synthesis that results in abnormal scar formation.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o871	List the three major classifications of connective tissue with their subclassification.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o872	Describe the two subtypes of embryonic connective tissue.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o873	Describe the two general subtypes of connective tissue proper.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o874	Describe the structure and location of lamina propria.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o875	List the types of specialized connective tissue.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o876	List common location sites for each type of specialized connective tissue.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o877	Compare and contrast white and brown adipose tissue in relation to its appearance, location and function.
			SOM.MK.1.BPM1.1.FTM.3.HCB.o878	List at least two commonly used stains to identify connective tissue under light microscopy
			SOM.MK.1.BPM1.1.FTM.3.HCB.o879	Describe the factors that lead to formation of interstitial fluid, the formation of lymph, and the development of edema.
Lecture 30	PHAR	Pharmacodynamics	SOM.MK.1.BPM1.1.FTM.4.PHAR.0001	Define the following terms: drug, pharmacology, pharmacodynamics, pharmacokinetics, pharmacotherapy and toxicology.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0002	Define drug receptor and list the different classes of receptors with which drugs interact.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0003	Give examples of drugs whose actions are not mediated by binding to receptors.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0004	Describe and interpret graded dose-response curves. Define E_{max} and EC_{50} .
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0005	Describe and interpret drug-receptor binding curves. Define K_D and B_{max} . Define affinity.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0006	Define spare receptors and signal amplification.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0007	Compare and contrast <i>in vitro</i> experiments with whole animal experiments.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0008	Define efficacy and potency. Indicate the positions on a graded dose-response curve that are used to define drug potency and efficacy.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0009	Define agonist, antagonist, full agonist, partial agonist, and inverse agonist.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0010	Describe the different mechanisms of receptor and nonreceptor antagonism.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0011	Describe and interpret quantal dose-response curves. Define ED_{50} , TD_{50} and LD_{50} .
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0012	Explain therapeutic index and therapeutic window.
Lecture 31	PHAR	Pharmacokinetics	SOM.MK.1.BPM1.1.FTM.4.PHAR.0013	Describe the mechanisms of drug permeation through cell membranes.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0014	Define drug absorption.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0015	Describe the main routes of drug administration. Explain the first-pass effect.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0016	Describe the factors that influence drug absorption.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0017	Explain the effect of pH on drug absorption. Describe and use the Henderson-Hasselbalch equation. Explain ion trapping.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0018	Explain why the intestine is the major site of drug absorption.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0019	Describe how P-glycoprotein affects drug absorption.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0020	Define bioavailability. Describe how bioavailability is determined.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0021	Define drug distribution. Describe the various factors that affect drug distribution.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0022	List the major mechanisms responsible for drug elimination.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0023	Describe how metabolism affects drug structure and function. List the main organs involved in drug metabolism.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0024	Describe the role of the cytochrome P450 enzymes in drug metabolism. Explain the clinical relevance of enzyme induction and enzyme inhibition. List the main drugs that can act as inhibitors or inducers of cytochrome P450 isoenzymes.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0025	Describe the main mechanisms of drug excretion.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0090	Differentiate between constitutive, inducible and repressible gene regulation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0091	Analyze the common component parts of general “operon” systems

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DLA 18	GNET	Prokaryotic Gene Regulation	SOM.MKIII.BPM1.1.FTM.3.GNET.0092	Identify the specific conditions by which individual operons will be transcribed: The influence of regulatory proteins which can activate or repress expression
			SOM.MKIII.BPM1.1.FTM.3.GNET.0093	Identify the specific conditions by which individual operons will be transcribed: Conditions which influence positive and negative regulation for each particular operon
			SOM.MKIII.BPM1.1.FTM.3.GNET.0094	Explain the significance of mutations within an operon and how they can affect gene regulation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0095	Compare & contrast eukaryotic & prokaryotic gene regulation (make your own chart)
Lecture 32	GNET	Gene Regulation	SOM.MKIII.BPM1.1.FTM.3.GNET.0096	Analyze cis-regulatory elements and trans-acting factors: Significance of different genes sharing common cis-regulatory elements
			SOM.MKIII.BPM1.1.FTM.3.GNET.0098	Discuss cell type specific gene expression
			SOM.MKIII.BPM1.1.FTM.3.GNET.0099	Explain how activator and repressor proteins interact with each other to regulate gene transcription
			SOM.MKIII.BPM1.1.FTM.3.GNET.0100	Define microRNA (miRNA); Compare and contrast regulation by miRNA to other types of gene regulation (make your own chart)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0101	Explain the various steps involved in the processing of miRNA
			SOM.MKIII.BPM1.1.FTM.3.GNET.0102	Describe how miRNA regulates gene expression
			SOM.MKIII.BPM1.1.FTM.3.GNET.0105	Explain the normal function of epigenetic gene silencing in development; including the role of X-inactivation as discussed Genomics and Human Variation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0107	Discuss how the environment can modify epigenetic gene regulation in a person and in subsequent generations
			SOM.MKIII.BPM1.1.FTM.3.GNET.0109	Define genomic imprinting
			SOM.MKIII.BPM1.1.FTM.3.GNET.0110	Demonstrate how genomic imprinting would appear in a human pedigree when either a male or female passes a mutation to their offspring
			SOM.MKIII.BPM1.1.FTM.3.GNET.0112	Outline examples of genomic imprinting and epigenetic syndromes including Angelman and Prader Willi discussed in Genomics and Human Variation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0113	Describe how 5-aza-2'-deoxycytidine (Decitabine; Dacogen) can be used to correct abnormal DNA methylation
Lecture 33	PHAR	Clinical Pharmacokinetics: Principles	SOM.MK.1.BPM1.1.FTM.4.PHAR.0026	Describe the aim of clinical pharmacokinetics.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0027	Define volume of distribution. Describe the factors that affect the volume of distribution of a drug.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0028	Explain the uses of the volume of distribution. Determine the volume of distribution of a drug. Recognize the distribution phase and the elimination phase in a plot of plasma drug concentration following an IV bolus.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0029	Define clearance. Distinguish between clearance and rate of drug elimination.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0030	Define first-order kinetics of drug elimination. Explain why most drugs follow first-order kinetics of elimination. Explain why clearance is constant for drugs that follow first-order kinetics of elimination.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0031	Define half-life. Calculate the drug plasma levels reached after a given number of half-lives during drug administration, and after drug administration is discontinued. Calculate the half-life of a drug graphically from a plot of the drug plasma level vs time. Describe the relationship between half-life, volume of distribution, and clearance.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0032	Describe the factors that affect half-life.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0033	Define zero-order kinetics of drug elimination. Explain why some drugs follow zero-order kinetics of elimination. List examples of drugs that follow zero-order kinetics of elimination.
Lecture 34	PHAR	Clinical Pharmacokinetics: Applications	SOM.MK.1.BPM1.1.FTM.4.PHAR.0034	Use pharmacokinetic concepts and principles to design a rational dosage regimen. Define maintenance dose. Describe the concept of steady-state with regard to plasma drug concentrations.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0035	Given a drug's clearance, calculate the maintenance dose needed to maintain an average steady-state drug level using either a continuous infusion or a fixed-dose, fixed-time interval regimen. Given a drug's clearance, and the dosing rate, calculate the steady-state drug plasma level achieved.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0036	Describe different dosage regimens.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0037	Define loading dose. Calculate a loading dose to achieve a target plasma drug level.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0038	Given the half-life of a drug, calculate the accumulation factor. Calculate peak and trough concentrations at steady state for a drug administered by multiple IV injections.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0039	Plot the time course of plasma drug concentration following different routes of administration. Compare and contrast the time course of plasma concentration for an immediate-release oral formulation and a slow-release oral formulation.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0887	List the functions of muscle tissue.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0888	Describe the structure of thick myofilaments in striated muscles.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0889	Identify the proteins associated with the thick myofilament in striated muscles.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0890	Give the function(s) of the proteins associated with the thick myofilament in striated muscles

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Lecture 35	HCB	Flipped Class: Histology of Muscle	SOM.MK.I.BPM1.1.FTM.3.HCB.0891	Describe the structure of thin myofilaments in striated muscles
			SOM.MK.I.BPM1.1.FTM.3.HCB.0892	Identify the proteins associated with the thin myofilament in striated muscles.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0893	Give the function(s) of the proteins associated with the thin myofilament in striated muscles
			SOM.MK.I.BPM1.1.FTM.3.HCB.0894	Give the function of the sarcomere.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0895	Describe the microscopic structure of the sarcomere.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0896	Describe the sliding filament model of muscle contraction.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0897	Describe briefly the events of the actomyosin cross-bridge cycle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0898	List the 2 types of striated muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0899	Describe the contractile apparatus of striated muscle fibers.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0900	State the functions of the accessory proteins typically associated with striated muscle cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0901	Give the function(s) of skeletal muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0902	State the localization of skeletal muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0903	Describe the microscopic structure of skeletal muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0904	Describe the general organization of skeletal muscle including connective tissue, muscle fascicles, muscle fibers, myofibrils and myofilaments.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0905	Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in skeletal muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0906	Describe the three types of skeletal muscle fibers.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0907	Discuss the regenerative capability of skeletal muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0908	Describe the ultrastructural defects in muscular dystrophies including Duchenne muscular dystrophy.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0909	Give the function of the muscle spindle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0910	Describe the histological structure of the muscle spindle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0911	Give the function of the Golgi tendon organs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0912	Describe the histological structure of the Golgi tendon organs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0913	Give the function(s) of cardiac muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0914	State the localization of cardiac muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0915	Describe the microscopic structure of cardiac muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0916	Describe the general organization of cardiac muscle, including the structure and function of intercalated discs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0917	Give the function(s) of Purkinje fibers
			SOM.MK.I.BPM1.1.FTM.3.HCB.0918	Describe the microscopic structure Purkinje fibers.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0919	State the localization of Purkinje fibers.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0920	Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in cardiac muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0921	Discuss the regenerative capability of cardiac muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0922	Give the function(s) of smooth muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0923	State the localization of smooth muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0924	Describe the microscopic structure of smooth muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0925	Describe the microscopic structure of the thick myofilaments in smooth muscles.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0926	Identify the proteins associated with the thick myofilament in smooth muscles.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0927	Give the function(s) of the proteins typically associated with the thick myofilament in smooth muscles
			SOM.MK.I.BPM1.1.FTM.3.HCB.0928	Describe the microscopic structure of thin myofilaments in smooth muscles
			SOM.MK.I.BPM1.1.FTM.3.HCB.0929	Identify the proteins associated with the thin myofilament in smooth muscles.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0930	Give the function(s) of the proteins typically associated with the thin myofilament in smooth muscles.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0931	Describe the contractile apparatus of smooth muscle cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0932	Describe the regenerative capability of smooth muscle.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0933	State the function(s) of the accessory proteins usually associated with smooth muscle cells.
DLA 19	ANAT	Introduction to the Nervous System 1 - General Overview	SOM.MK.II.BPM1.1.FTM.1.ANAT.1004	Describe the three basic functions of the nervous system.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1005	Describe the anatomical & functional divisions of the nervous system.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1006	Describe the components of the central nervous system (CNS).
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1007	Describe the components of the peripheral nervous system (PNS).
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1008	Describe the components and targets of the somatic nervous system.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1009	Describe the components of the autonomic nervous system.
Reading Assignment	HCB	Nerve	SOM.MK.I.BPM1.1.FTM.3.HCB.0936	Describe the following neurotransmitters: acetylcholine, norepinephrine and epinephrine.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0952	Describe the two types of axonal transport (anterograde vs. retrograde).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0934	Describe the structure of a typical synapse.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0935	Compare chemical vs. electrical synapse.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0937	Describe the structure of a typical neuron using a multipolar neuron as an example
			SOM.MK.I.BPM1.1.FTM.3.HCB.0938	State the function of the neuromuscular junction.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0939	Describe the histological structure of the neuromuscular junction.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0940	Define a nerve
			SOM.MK.I.BPM1.1.FTM.3.HCB.0941	Define a tract
			SOM.MK.I.BPM1.1.FTM.3.HCB.0942	Differentiate between nerve vs. tract based on location.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0943	State the contents of a ganglion.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 36	HCB	Histology of Nerve Tissue - Peripheral Nervous System	SOM.MK.I.BPM1.1.FTM.3.HCB.0944	State the contents of a nucleus.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0945	Differentiate between ganglia vs. nuclei based on location.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0946	Identify sensory ganglia based on their location.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0947	Describe the microscopic structure of a sensory ganglia using the Dorsal Root Ganglion (DRG) as an example.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0948	Identify the two autonomic ganglia based on location.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0949	Describe the microscopic structure of the sympathetic autonomic ganglia using the sympathetic chain as an example.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0950	Describe the microscopic structure of the parasympathetic autonomic ganglia using the myenteric plexus as an example.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0951	Describe the structure, locations and function (s) of the three general categories of neurons (sensory, interneurons, motor).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0953	List the function(s) of the peripheral neuroglia (Schwann and capsular cells).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0954	Describe the microanatomical structure of the peripheral neuroglia (Schwann and capsular cells).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0955	Explain the process of myelination noting differences between Oligodendrocytes and Schwann Cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0956	Compare myelinated vs. unmyelinated nerves.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0957	Discuss Guillain-Barre as a demyelinating disease in the PNS
			SOM.MK.I.BPM1.1.FTM.3.HCB.0958	Describe the connective tissue sheaths of a peripheral nerve.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0959	State the function(s) of blood-nerve barrier
			SOM.MK.I.BPM1.1.FTM.3.HCB.0960	Describe the microscopic structure of the blood-nerve barrier including the location and cells that form it
DLA 20	ANAT	Introduction to the Nervous System 2 - CNS and PNS	SOM.MK.II.BPM1.1.FTM.1.ANAT.1011	Describe the basic organization and function of the brain and cranial nerves
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1012	Describe the basic organization and function of the spinal cord and spinal nerves
Lecture 37	HCB	Histology of Nerve Tissue - Central Nervous System	SOM.MK.I.BPM1.1.FTM.3.HCB.0961	Give the function(s) of the choroid plexus.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0962	Describe the histological structure of the choroid plexus.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0963	Describe the general organization of white and gray matter in the brain.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0964	Describe the microscopic structure of the cerebral cortex, including characteristics of the pyramidal cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0965	Define "neuropil".
			SOM.MK.I.BPM1.1.FTM.3.HCB.0966	Describe the microscopic structure of the cerebellar cortex.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0967	Identify Purkinje cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0968	Describe the general organization of white and gray matter in the spinal cord.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0969	Describe the histological appearance of the white matter of the spinal cord.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0970	Describe the histological appearance of the gray matter of the spinal cord, including characteristics of alpha motor neurons.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0971	Differentiate between the three meningeal layers based on structure, function(s) and location.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0972	List the functions of the central neuroglia including astrocytes, oligodendrocytes, microglia and ependymal cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0973	Describe the microanatomical features of the structure of the central neuroglia including astrocytes, oligodendrocytes, microglia and ependymal cells.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0974	Compare CNS vs PNS responses to neuron injury
			SOM.MK.I.BPM1.1.FTM.3.HCB.0975	State the function of the blood-brain barrier.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0976	Describe the histological structure of the blood-brain barrier.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0977	State the function of the blood-CSF barrier.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0978	Describe the histological structure of the blood-CSF barrier.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0979	Discuss Multiple Sclerosis as a demyelinating disease in the CNS
DLA 21	ANAT	Anatomy of the Autonomic System	SOM.MK.II.BPM1.1.FTM.1.ANAT.1013	List the three target tissues of visceromotor (ANS) fibers.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1014	Define the terms preganglionic and postganglionic.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1015	List the four cranial nerves that contain preganglionic parasympathetic fibers.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1016	List the segments of the spinal cord that contain the neural cell bodies of preganglionic parasympathetic fibers.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1017	List the locations of the neural cell bodies of the postganglionic parasympathetic fibers
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1018	Identify the segments of the spinal cord that contain the neural cell bodies of preganglionic sympathetic fibers and use this information to interpret clinical signs and symptoms
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1019	List the locations of the neural cell bodies of the postganglionic sympathetic fibers
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1020	Describe the location and extent of the sympathetic trunk.
Lecture 38	PHYS	Physiology of Autonomic Transmissions	SOM.MK.II.BPM1.1.FTM.3.PHYS.0040	Describe the main functions of the SNS and the PNS.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0041	Classify autonomic nerves based on the primary neurotransmitters released from their terminals.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0043	Describe the processes involved in cholinergic and adrenergic transmission.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0044	Describe the location and function of cholinergic and adrenergic receptor subtypes and the signaling mechanisms associated with them.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0045	Explain the concepts of autonomic tone, dual innervation, and predominant tone in the target tissues of the autonomic nervous system.
Lecture 39	PHYS	Flipped Class: Autonomic Integration	SOM.MK.II.BPM1.1.FTM.3.PHYS.0046	Explain the responses of end organs to stimulation of sympathetic and parasympathetic divisions of the autonomic nervous system
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0047	List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0048	List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.
Lecture 40	PHAR	Cholinergic Agonists and Antagonists	SOM.MK.1.BPM1.1.FTM.4.PHAR.0040	List the different classes of cholinergic drugs.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0041	Describe the muscarinic and nicotinic actions of acetylcholine.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0042	Describe the two classes of direct-acting cholinergic agonists. Describe the organ system effects, mechanism of action, uses and adverse effects of muscarinic receptor agonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0043	Describe the organ system effects, mechanism of action, uses and adverse effects of nicotinic receptor agonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0044	Describe the organ system effects, mechanism of action, uses and adverse effects of indirect-acting cholinergic agonists (anticholinesterases). Compare and contrast the effects of direct-acting and indirect-acting cholinergic agonists on the cardiovascular system.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0045	List the different classes of cholinergic antagonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0046	Describe the organ system effects, mechanism of action, therapeutic uses, adverse effects and contraindications of muscarinic receptor antagonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0047	Describe the mechanism of action of the ganglion blockers and predict their effects by applying knowledge of the predominant tone of a dually-innervated organ.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0048	Describe the mechanism of action, uses and main adverse effects of the nondepolarizing and the depolarizing neuromuscular blockers.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0049	Describe the mechanism of action and uses of botulinum toxin.
Lecture 41	BCHM	Purine Nucleotide Synthesis and Salvage Pathway	SOM.MK.I.BPM1.1.FTM.3.BCHM.0020	Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0084	Describe the role of thiopurine methyl transferase (TPMT) in the conversion of 6-mercaptopurine to an inactive product.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0085	Explain the mechanism of 6-mercaptopurine/azathioprine toxicity in patients who are homozygous for autosomal recessive polymorphisms in the TPMT gene (example of an adverse drug reaction).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0162	Enumerate C and N donors of purine ring (amino acids and 1-C groups)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0163	Outline denovo purine biosynthesis and indicate the role of pentose phosphate pathway as donors of pentoses (PRPP). Review formation of AMP and GMP from IMP
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0164	Discuss the regulation of denovo purine biosynthesis (Identify the regulatory enzyme and positive and negative modulators)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0165	Correlate mechanism of action of sulfa drugs, trimethoprim, methotrexate, mycophenolic acid to their clinical application
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0166	Explain formation of deoxyribonucleotides from ribonucleotides (Specify coenzyme and inhibitor)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0167	Identify reactions catalyzed by HGPRT and APRT
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0168	Correlate clinical, laboratory (hyperuricemia) and molecular findings in Lesch-Nyhan syndrome (Specify enzyme deficient).
Lecture 42	BCHM	Purine Degradation and Pyrimidine Metabolism	SOM.MK.I.BPM1.1.FTM.3.BCHM.0169	Analyze causes of hyperuricemia (overproduction and underexcretion of uric acid)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0170	Analyze causes of hyperuricemia (overproduction and underexcretion of uric acid)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0171	Correlate the basis of clinical features and explain basis of use of allopurinol and febuxostat in the management of hyperuricemia
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0172	Propose molecular basis of immunodeficiency in autosomal recessive SCID (ADA deficiency) and deficiency of purine nucleoside phosphorylase.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0173	Enumerate the donors of C and N atoms to pyrimidine ring
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0174	Outline pyrimidine biosynthesis (UTP, CTP and dTMP). Distinguish purine and pyrimidine biosynthetic pathways
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0175	Summarize the regulation of pyrimidine biosynthesis (regulatory enzymes, regulators)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0176	Explain formation of deoxyribonucleotides from ribonucleotides (Specify coenzyme and inhibitor)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0177	Indicate mechanism of action of 5-fluoro uracil and methotrexate and their clinical application
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0178	Specify enzyme deficiency and explain metabolic basis of clinical features in orotic aciduria
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0195	Compare and contrast the roles and subcellular location of CPS-I (urea cycle) and CPS-II (pyrimidine biosynthesis)
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0050	Describe the physiological basis of adrenergic responses.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0051	List the different classes of adrenergic agonists.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 43	PHAR	Adrenergic Agonists and Antagonists	SOM.MK.1.BPM1.1.FTM.4.PHAR.0052	Describe the pharmacological effects, uses and adverse effects of the endogenous catecholamines.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0053	Contrast the cardiovascular effects of a low dose and a high dose of epinephrine.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0054	Explain the baroreceptor reflex.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0055	Describe the pharmacological effects, uses and adverse effects of beta agonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0056	Describe the pharmacological effects, uses and adverse effects of alpha1 and alpha2 agonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0057	Describe the mechanism of action, pharmacological effects, uses and adverse effects of the indirect-acting and mixed-action adrenergic agonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0058	List the different classes of adrenergic antagonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0059	Describe the pharmacological effects, uses and adverse effects of alpha adrenergic antagonists.
			SOM.MK.1.BPM1.1.FTM.4.PHAR.0060	Describe the pharmacological effects, uses and adverse effects of beta adrenergic antagonists. Compare and contrast the adverse effects of the nonspecific and cardiospecific beta adrenergic antagonists
Lecture 44	GNET	Multifactorial Inheritance	SOM.MKIII.BPM1.1.FTM.3.GNET.0114	Discuss quantitative traits in terms of multifactorial inheritance
			SOM.MKIII.BPM1.1.FTM.3.GNET.0115	Analyze how quantitative traits can be controlled by many genes each of which might have minimal contribution to the phenotype
			SOM.MKIII.BPM1.1.FTM.3.GNET.0116	Discuss general concepts of multifactorial disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0117	Analyze the threshold model of multifactorial inheritance
			SOM.MKIII.BPM1.1.FTM.3.GNET.0118	Analyze heritability of multifactorial disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0119	Discuss methods used to identify genetic or environmental risk factors and contributors for multifactorial disorders
Lecture 45	GNET	Karyotypes and Aneuploidy	SOM.MKIII.BPM1.1.FTM.3.GNET.0120	Analyze the normal human karyotype and how it is obtained
			SOM.MKIII.BPM1.1.FTM.3.GNET.0121	Analyze the abnormal human karyotype and define the terms euploidy and aneuploidy. Use disorders and syndromes such as Down, Patau, Edward, Turner, Klinefelter as examples
			SOM.MKIII.BPM1.1.FTM.3.GNET.0122	Analyze meiosis to explain how chromosomal aberrations are formed
			SOM.MKIII.BPM1.1.FTM.3.GNET.0123	Discuss mechanisms that might lead to mosaicism for chromosomal aneuploid conditions
Lecture 46	GNET	Structural Chromosomal Abnormalities	SOM.MKIII.BPM1.1.FTM.3.GNET.0124	Analyze balanced, unbalanced, reciprocal and Robertsonian translocations, describe how a reciprocal translocation can create a fusion gene
			SOM.MKIII.BPM1.1.FTM.3.GNET.0125	Describe how the Robertsonian translocation can explain Down syndrome. Indicate the risk that a carrier of a Robertsonian translocation has for having a child with Down syndrome
			SOM.MKIII.BPM1.1.FTM.3.GNET.0126	Analyze microdeletion syndromes with the following examples: cri-du-chat, 22q11.2 DS, Wolf_Hirschhorn, Prader-Willi, Angelman
			SOM.MKIII.BPM1.1.FTM.3.GNET.0127	Compare and contrast other chromosomal abnormalities including inversions, ring chromosomes, uniparental disomy and isochromosomes
			SOM.MKIII.BPM1.1.FTM.3.GNET.0128	Analyze cytogenomic tests used for diagnosis of chromosomal aberrations: Karyotype, FISH, Microarray CGH
Lecture 47	GNET	Genetics of oncogenes	SOM.MKIII.BPM1.1.FTM.3.GNET.0130	Define and differentiate between a protooncogene and an oncogene. Summarize the various ways in which oncogenes may transform cells
			SOM.MKIII.BPM1.1.FTM.3.GNET.0131	Compare and contrast tumor suppressor genes, DNA repair genes, and oncogenes
			SOM.MKIII.BPM1.1.FTM.3.GNET.0132	Identify the different classes and functionality of proteins involved in the major cellular proliferative pathway and how mutations in which can result in cancer which includes: Growth factors (hormones), growth factor receptors
			SOM.MKIII.BPM1.1.FTM.3.GNET.0133	Identify the different types of mutation that would result in the activation of a protooncogene to an oncogene such as point mutation, nonsense mutation, chromosomal translocation, and gene amplification; provide examples for each)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0134	Indicate the specific chromosomal translocation in chronic myeloid leukemia and Burkitt lymphoma. Indicate the genes that are affected by the translocation in each cancer and the molecular basis of tumorigenesis in each example. Interpret a cytogenetic test that detects translocations
			SOM.MKIII.BPM1.1.FTM.3.GNET.0135	Relate the pharmaceutical application of the molecular mechanisms of cancer to the development of anticancer drugs. Include the following specific examples: Herceptin in HER2+ breast cancer, Imaniteb mesylate for Ph+ chronic myeloid leukemia
			SOM.MKIII.BPM1.1.FTM.3.GNET.0136	Compare and contrast the function of the following common oncogenes: Receptor Tyrosine Kinase, Ras, transcription factors
			SOM.MKIII.BPM1.1.FTM.3.GNET.0137	Comprehend the progressive and cumulative nature of cancer at the genetic level using the example of FAP - Compare and contrast FAP to HNPCC (Lynch syndrome)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0138	Compare and contrast the function of the following common tumor suppressor genes: RB, p53, Wilms tumor gene, APC
			SOM.MKIII.BPM1.1.FTM.3.GNET.0139	Distinguish the role of tumor suppressor genes including: mutator genes (DNA repair; caretakers); genes involved in cell cycle regulation (gatekeepers); and genes involved in apoptosis regulation

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 48	GNET	Genetics of Tumor Suppressor Genes	SOM.MKIII.BPM1.1.FTM.3.GNET.0140	Indicate the genetic basis of Li-Fraumeni syndrome
			SOM.MKIII.BPM1.1.FTM.3.GNET.0141	Analyze the two-hit model where loss of heterozygosity for the action of tumor suppressor genes (inherited mutation vs. sporadic) to understand how the inheritance pattern differs between sporadic and inherited Rb
			SOM.MKIII.BPM1.1.FTM.3.GNET.0142	Interpret familial cancer syndromes and their genetic principles to describe how a predisposition to cancer can be transmitted to offspring use BRCA and APC as examples.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0143	Determine the characteristics of inherited colorectal (familial adenomatous polyposis and hereditary nonpolyposis colon cancer) and breast cancers
			SOM.MKIII.BPM1.1.FTM.3.GNET.0144	Differentiate the clinical presentation of inherited cancers from that of sporadic cancers
			SOM.MKIII.BPM1.1.FTM.3.GNET.0145	Indicate the utility of expression array analysis in cancer
			SOM.MKIII.BPM1.1.FTM.3.GNET.0146	Identify the role of epigenetic changes and microRNA in tumorigenesis
Lecture 49	GNET	DNA Repair	SOM.MKIII.BPM1.1.FTM.3.GNET.0147	Analyze ways that DNA can be damaged, the cellular mechanisms that are employed in an attempt to fix the damage, and the long and short term consequences of mutation accumulation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0148	Analyze genetic disorders associated with defects of DNA repair including Bloom syndrome, Lynch syndrome (HNPCC), xeroderma pigmentosa, ataxia telangiectasia, BRCA 1/2 hereditary breast and ovarian cancer
			SOM.MKIII.BPM1.1.FTM.3.GNET.0149	Discuss why some areas of the genome are more prone to mutation than other areas; such as repetitive regions, low copy repeats, and 5-methyl cytosine
			SOM.MK.III.BPM1.1.FTM.3.GNET.0150	Discuss the central role of p53 in regulation of cell proliferation, cell survival, and apoptosis
			SOM.MK.III.BPM1.1.FTM.3.GNET.0151	Discuss the role of BAX and BCL-2 in regulating apoptosis: -BAX as pro-apoptotic, -BCL-2 as anti-apoptotic
			SOM.MK.III.BPM1.1.FTM.3.GNET.0152	Indicate how cytochrome c release from mitochondria may initiate apoptosis
			SOM.MK.III.BPM1.1.FTM.3.GNET.0153	Summarize the extrinsic, intrinsic, and perforin pathway of apoptosis with respect to cancer.
Lecture 50	BCHM	Flipped Class: Nutrition in Health and Disease	SOM.MK.III.BPM1.1.FTM.3.BCHM.0183	Summarize concept of energy equilibrium: positive and negative energy balance with examples.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0184	Analyze components of daily energy expenditure: a. Resting energy requirement (RMR/BMR) b. Physical activity c. Diet induced thermogenesis.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0185	Indicate macronutrient ratios of a balanced diet. List dietary mono-, di- and polysaccharides.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0186	Describe relationship between dietary lipids and serum cholesterol in healthy individuals.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0187	Highlight significance of macronutrients and micronutrients (vitamins and minerals).
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0188	List energy available from macronutrients including ethanol. Calculate energy available from macronutrients if dietary data is provided.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0189	Describe significance of dietary lipids including cholesterol and fatty acids. Identify beneficial effects of supplementation with fish oil: EPA and DHA.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0004	Discuss importance of dietary essential fatty acids: linoleic acid (omega-6) and alpha-linolenic acid (omega-3)
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0190	Compare Mediterranean diet and Western diet.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0191	Indicate significance of dietary carbohydrate. Determine significance of glycemic index. Review composition of high fructose corn syrup.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0192	Explore significance of dietary fiber and correlate health benefits of dietary fiber.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0193	Analyze significance of dietary protein and list the dietary essential amino acids. Differentiate between sources of high and low quality protein. Review the significance of combining wheat and beans to improve protein biological value.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0195	Define states of positive and negative nitrogen balance with examples for each. Highlight changes in tissue protein in these states.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0196	Define Respiratory Quotient (RQ). Identify RQ in a mixed diet and during starvation.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0197	Explore roles of ghrelin, cholecystokinin, insulin and leptin involved in appetite regulation.
			SOM.MK.III.BPM1.1.FTM.3.BCHM.0199	Describe and review significance of indicators of nutritional status: a. BMI b. Estimation of lean body mass versus adipose tissue. c. Biochemical indicators: Serum albumin